

CHANGING PERSPECTIVES

Artist Sue Austin performs “Creating the Spectacle” in the Red Sea. Austin is changing people’s perspectives about wheelchairs.



GOALS

IN THIS UNIT, YOU WILL:

- Read about someone who thinks in pictures.
- Learn about someone who challenges the way people think about her disability.
- Explore how disability contributes to people's experiences.

THINK AND DISCUSS

1. Look at the photo. How do you think this person feels?
2. What surprises you about the photo?

Lesson A

PRE-READING

- A.** Autism (sometimes called autism spectrum disorder, or ASD) affects how someone experiences and interacts with the world. What do you know about autism? Discuss with a partner.

Answers will vary. See Teaching Notes for a detailed explanation of autism.

- B.** Look at the photo and skim the first two paragraphs on this page. What do you think this passage is mostly about?

- a. A woman who studies the challenges of certain disabilities
- b. A woman who teaches people about her disability
- ☒ c. A woman who has turned her disability into an advantage

C. Write your answers to the questions below.

1. What possible challenges might people face when working with animals?

Answers will vary. Animals can be unpredictable, and sometimes even dangerous. It may be hard for most people to understand how animals think and why they behave in certain ways.

2. What personal qualities do you think a person should have to work well with animals?

Answers will vary.



THINKING IN PICTURES

Temple Grandin designs plans for livestock farms that reduce the fear that cattle may experience.



THINKING IN PICTURES

We often make incorrect assumptions about people when we meet them for the first time. This may be especially true when that person has a disability or an illness. Temple Grandin challenges these assumptions.

Temple Grandin is an expert in animal behavior. She also has autism. Some people view autism as a very limiting disability, but Grandin disagrees. She believes that if there were no autism in the world, we would “still be socializing in front of a wood fire at the entrance to a cave.” Why? Grandin thinks that some people with autism have brains that can solve problems “normal” brains cannot.



No one knows for sure what causes autism, but scientists believe it begins during the very early stages of brain development. Autism exists on a spectrum—from very **severe** to moderate—and it **affects** people in different ways. Often, people with autism have trouble socializing and communicating normally with other people. For Grandin, autism also affects the way she thinks and understands the world.

VISUAL THINKER

Grandin, like many people with autism, is a visual thinker—“I think in pictures, not in language,” she says. For example, if you ask most people to think of a shoe, they might “see” in their mind one or two shoes. Grandin, however, remembers all the **specific** shoes that she knows and visualizes them in her mind. She says that her mind “works like Google Images.” Like Google, she can very quickly categorize the images and **arrange** them in a specific order. She can also “animate” the images and make them into little movies.



This ability to think **visually** has helped Grandin in her career. For example, she has a unique understanding of how cattle think. According to animal researchers, cattle think in pictures, not language—similar to how Grandin’s mind works. With this insight, she was able to design plans for livestock farms that reduce the fear that cattle may experience. For example, cattle are easily distracted and worried by things like waving flags or even a hose on the ground. Grandin’s designs take away those visual distractions, resulting in calmer livestock.

Grandin’s visual thinking means she can also picture how her designs might work. She creates a mental **simulation** of her designs and does a test run of the equipment in her mind. She says that the “normal brain **ignores** the details,” but her brain looks at every tiny aspect of the design. She makes sure the system works even before it’s built.



According to Grandin, an autistic mind is “a specialist mind—good at one thing, bad at something else.” Her autism has made her an expert in animal **welfare**. She believes that other people with autism may be able to solve global problems or invent some incredible software. As she points out, “the world needs all kinds of minds.”

assumptions: *n.* things that you believe are true even though you have no proof

spectrum: *n.* a range of qualities, situations, etc.

distracted: *n.* turn your attention away from something you want to concentrate on

AUTISM SPECTRUM

Autism Spectrum Disorder (ASD) affects individuals in different ways and can range from very mild to severe.

ASD
is a neurological
condition that affects
the way a person
experiences and
interacts with
the world.



delayed & disordered
language



impaired social
interaction



repetitive
behavior

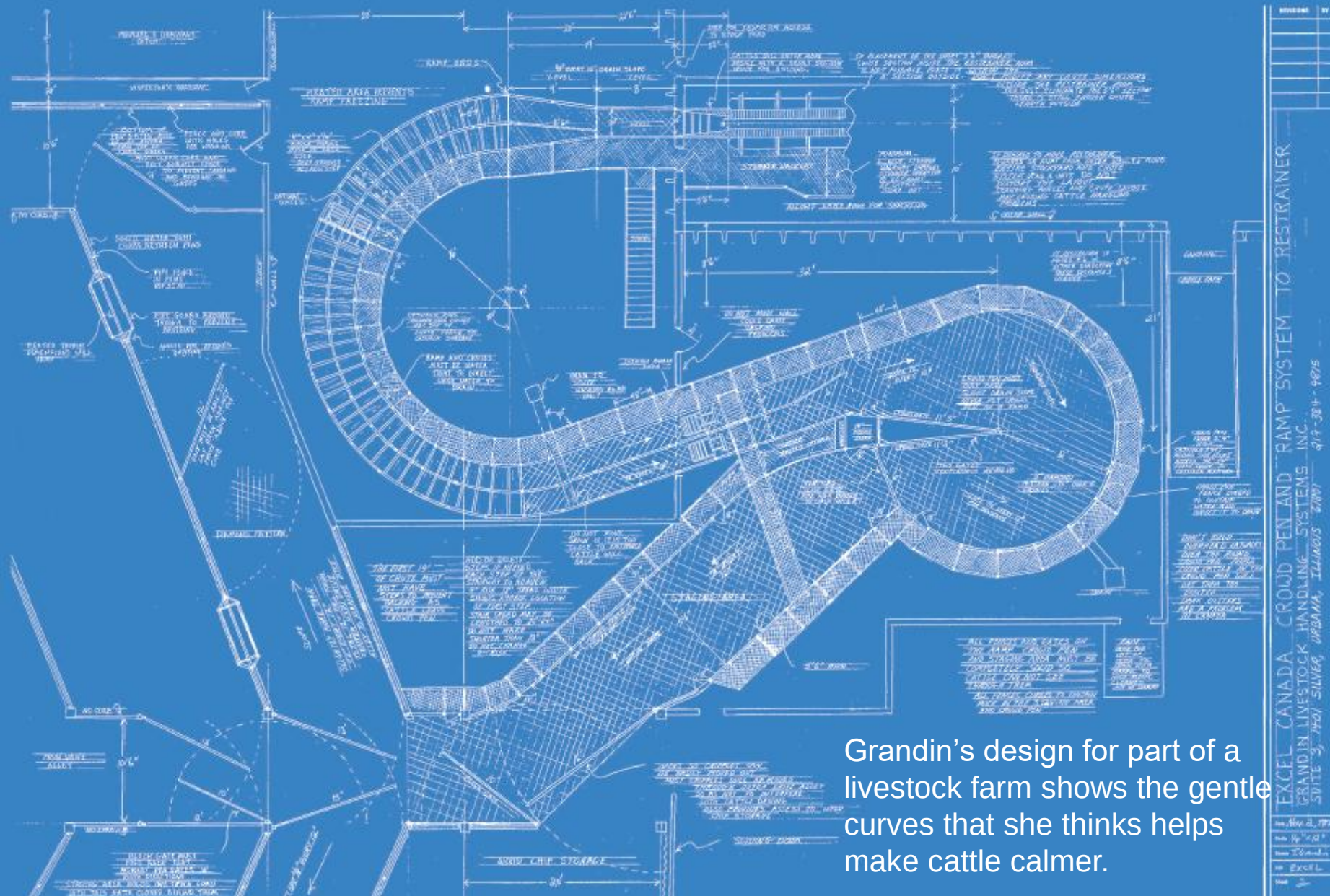


restricted range
of interest

1 *in* 50 KIDS HAVE
ASD
IN THE U.S.



BOYS
ARE **4x** MORE LIKELY
TO HAVE
ASD



Grandin's design for part of a livestock farm shows the gentle curves that she thinks helps make cattle calmer.

Developing Reading Skills

SCANNING FOR INFORMATION

Answer the questions below about the passage and infographic on pages 122–124.

1. What causes autism?

The cause of autism is unknown.

2. According to scientists, when does autism begin?

Scientists believe it begins in the early stages of brain development.

3. Who is more likely to have autism, boys or girls?

Boys are four times more likely than girls to have autism.

4. What are three effects of autism?

A person with autism may have delayed and disordered language, impaired social interaction, repetitive behavior, and/or a restricted range of interests.

GETTING THE MAIN IDEAS

Use information from the passage on pages 122–124 to answer each question.

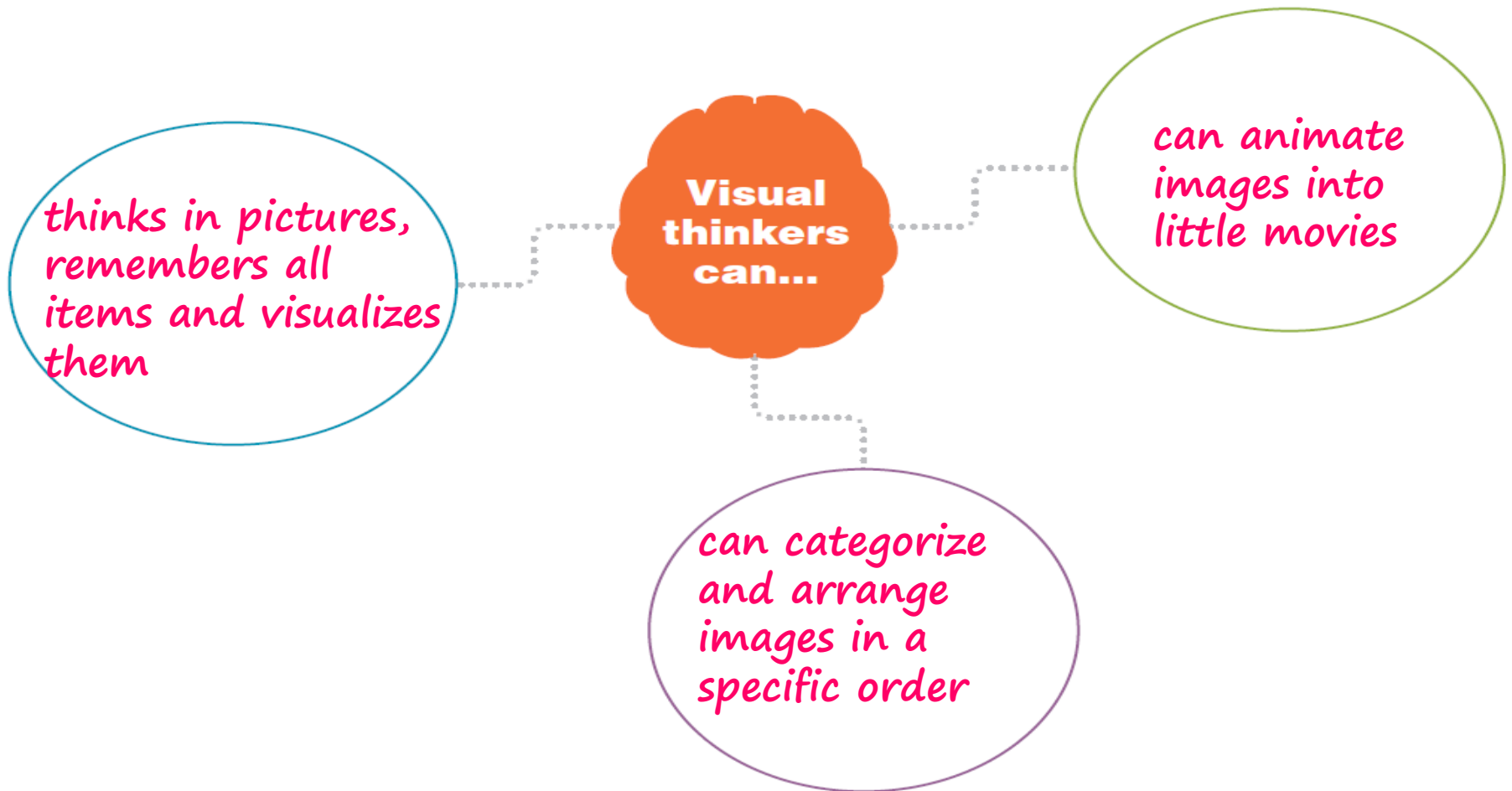
1. What does Temple Grandin do?
 - ☒ a. She works with animals.
 - b. She works with autistic children.
 - c. She is a farmer.
2. What does Temple Grandin say about autism?
 - a. Autism makes a person good at a lot of different things.
 - ☒ b. People with autism have specialized skills.
 - c. We have to find a cure for autism.
3. According to Grandin, how does her autism help her in her job? Complete the sentence.

Autism lets Grandin _____.

 - a. avoid using language to communicate
 - ☒ b. think in pictures in a similar way to animals
 - c. talk more easily to animals

UNDERSTANDING KEY DETAILS

Grandin describes herself as a visual thinker. Complete the chart with three things her mind is able to do according to the passage on pages 122–124.



BUILDING VOCABULARY

A. Complete the paragraph using the words below.

affect

visually

simulations

welfare

Temple Grandin understands how simple things like dark shadows and waving flags affect animals. When she designs her systems for farms and cattle-processing plants, she works hard to ensure that animals don't see or encounter things that will make them feel afraid. Grandin can visualize her plans in her mind. These mental simulations help her understand how her systems might work. Due to her ability to think visually and understand animals, Grandin has important insights into animal welfare.

B. For each bold word, choose the word or phrase that is closer in meaning.

b 1. **ignore**

a. notice

b. pay no attention to

a 2. **arrange**

a. sort

b. understand

b 3. **specific**

a. general

b. particular

a 4. **severe**

a. strong

b. moderate

GETTING MEANING FROM CONTEXT

According to the passage, Grandin does a “test run” of her designs in her mind. What is a test run? Do you ever do test runs of things in your mind? Note some examples.

It's a practice try or trial run to test something out before putting it to use.

CRITICAL THINKING

1. **Inferring.** Grandin believes that if there were no autism in the world, we would “still be socializing in front of a wood fire at the entrance to a cave.” What do you think she means by this?

She means that autistic thinkers have helped solve problems that have pushed human progress forward.

2. **Reflecting.** How do you think Grandin’s story might help change people’s perceptions of autism? Discuss with a partner.

Answers will vary. She presents individuals with autism as people with unique skills that are very useful, instead of as people with a disability.

EXPLORE MORE

Watch Temple Grandin’s TED Talk called “The world needs all kinds of minds” at TED.com. What other things did you learn about autism that surprised you? Share what you learned with the class.

Lesson B TED TALKS

DEEP SEA DIVING . . . IN A WHEELCHAIR



SUE AUSTIN Performance artist, TED speaker

When artist Sue Austin was a young woman, a serious illness forced her to start using a wheelchair.

When she began using the wheelchair, she noticed that some people behaved differently toward her. She felt they saw her mainly as a person with limited mobility, without taking the time to appreciate her abilities and interests. Today, she creates art to try to change people's preconceptions about disabilities.

One way that Austin changes people's ideas is through an artistic organization called Freewheeling. In one project, Austin put paint on the wheels of her wheelchair and drew lines on the sidewalk. These loops and wavy lines show the freedom that her wheelchair gives her.

preconceptions: *n.* opinions that people form about something before they have a lot of information about it

In this lesson, you are going to watch segments of Austin's TED Talk. Use the information above to answer the questions.

1. Why did Sue Austin begin using a wheelchair?

due to a serious illness

2. How did people behave toward Austin when she started using the wheelchair?

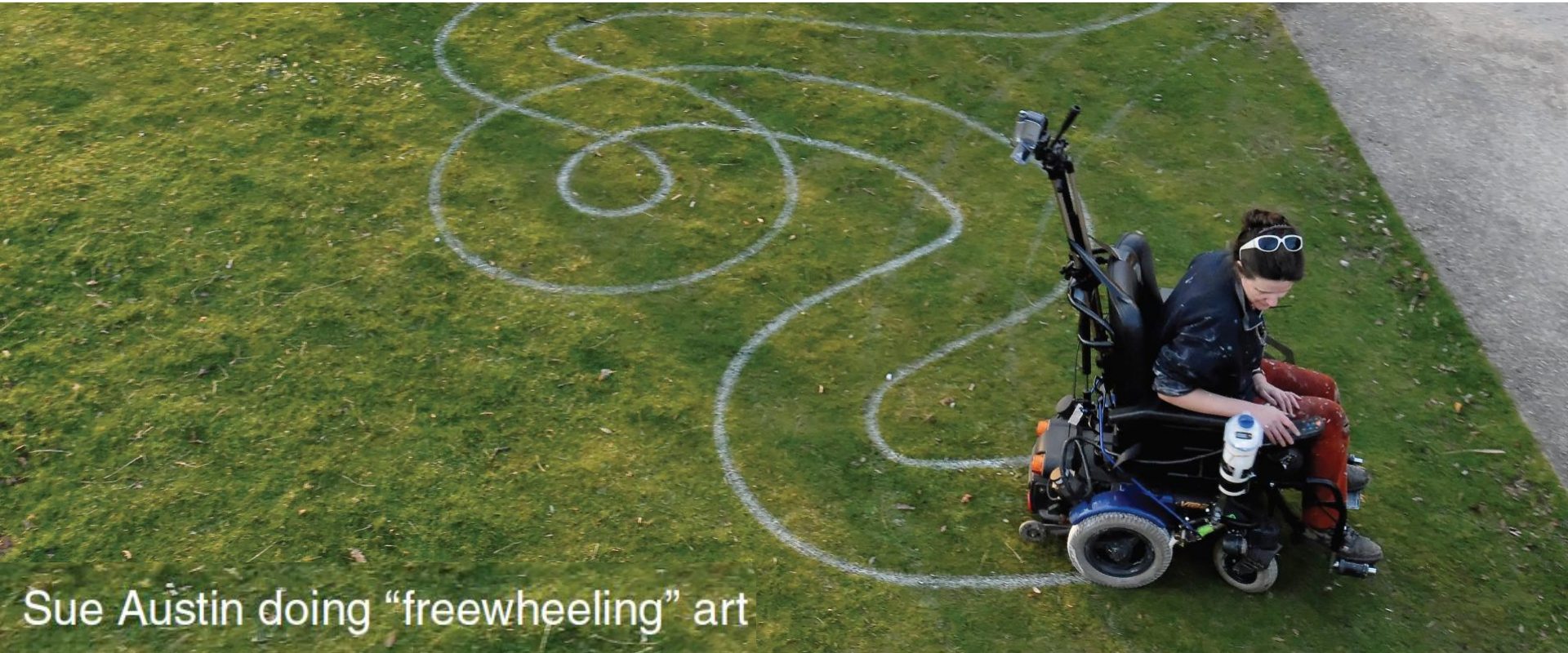
She felt they saw her as a person with limited mobility, and therefore didn't take time to get to know her.

3. How do you think Austin's project with Freewheeling can change the way people see disabilities?

Answers will vary.

Sue Austin's **idea worth spreading** is that a wheelchair doesn't have to mean "disability." It can be a new way to see and experience the world.





Sue Austin doing “freewheeling” art

PART 1

CHANGING IDENTITIES

PREVIEWING

- A.** Read the excerpt below from Sue Austin's talk. Complete the excerpt with your own ideas. Then share your ideas with a partner.

Guesses will vary. Actual answers are:

"When I started using the wheelchair, it was a tremendous new freedom. I'd seen my life slip away and become restricted. It was like having an enormous new toy . . . But even though I had this newfound joy and freedom, people's reaction completely changed towards me. It was as if they couldn't see me anymore, as if an invisibility cloak had descended. They seemed to see me in terms of their assumptions of what it must be like to be in a wheelchair."

tremendous: *adj.* extremely good

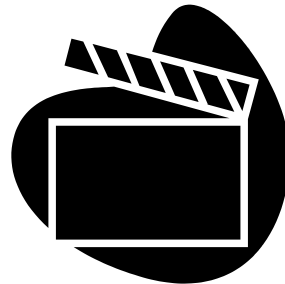
invisibility cloak: *n.* a cape that covers a person and makes it impossible for other people to see him or her

- B.** How do you think Austin felt when she started using the wheelchair?
Discuss with a partner.

Austin felt happy and excited to be able to move around again.

GETTING THE MAIN IDEAS

- A.** Watch (▶) the first segment of Austin's TED Talk and check your answers to the Previewing activities.



B. Read the excerpt below from Austin's talk and then answer the question.

“When I asked people their associations with the wheelchair, they used words like *limitation*, *fear*, *pity*, and *restriction*. I realized I'd internalized these responses and it had [changed] who I was on a core level. A part of me had become alienated from myself. I was seeing myself not from my perspective, but vividly and continuously from the perspective of other people's responses to me.”

internalize: v. to make a belief or value part of your way of thinking

alienate: v. to make a person feel like he or she is no longer part of a whole

How did Austin respond to people's reactions?

She explains that she began to believe their reactions and feel as if she really was limited and restricted by the wheelchair. The reactions began to negatively affect her perception of herself.

CRITICAL THINKING

Inferring. Austin says that she wants to “reclaim her identity.” What does she mean by this? Discuss with a partner.

She means she wanted to stop being influenced by other people's limiting views (of people in wheelchairs) and reconnect with who she really is as an individual.

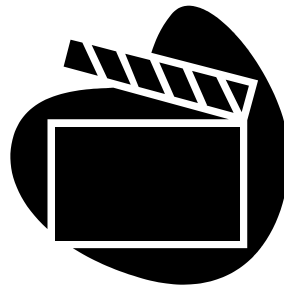
PART 2

AN AMAZING JOURNEY

UNDERSTANDING MAIN IDEAS

- A. Watch (▶) the next segment of Austin's talk. Write four words or phrases that describe how you feel when you watch her underwater dive.

Answers will vary.



B. Complete the sentences about Austin.

1. Austin expresses the _____ that she feels in her wheelchair.
 - ☒ a. joy and freedom
 - b. creativity and curiosity
 - c. satisfaction and relief

2. According to Austin, when people see her underwater in her wheelchair, they _____.
 - a. feel surprised or confused
 - b. feel nervous for her
 - ☒ c. feel the same way she does

RECOGNIZING TONE AND MESSAGE

Read the excerpt from Austin's talk. Then answer the questions below.

“It is the most amazing experience, beyond most other things I’ve experienced in life. I literally have the freedom to move in 360 degrees of space and an ecstatic experience of joy and freedom. And the incredibly unexpected thing is that other people seem to see and feel that, too . . .

And I’m thinking, it’s because in that moment . . . they have to think in a completely new way. And I think that moment of completely new thought perhaps creates a freedom that spreads to the rest of other people’s lives. For me, this means that they’re seeing the value of difference, the joy it brings when, instead of focusing on loss or limitation, we see and discover the power and joy of seeing the world from exciting new perspectives..”

1. Why does Austin think people's reactions to her wheelchair changed?

Seeing her in the chair underwater gives people a new way to view wheelchairs. They see being in a wheelchair as a valuable and different experience, not one related to limitation and loss.

2. Which two statements do you think best match Austin's overall message?

Discuss with a partner.

- a. People with disabilities face challenges that most people are not aware of.
- ☒ b. People should value their differences and not see just limitations and restrictions.
- ☒ c. Seeing or experiencing something for the first time can create a sense of freedom.

CRITICAL THINKING

Comparing. What perceptions did Temple Grandin and Sue Austin want to change? What did Grandin and Austin do to change people's perceptions?

Both Grandin and Austin are challenging the standard thinking that a physical or mental disability is a kind of limitation. Instead, both women show us how disabilities actually create an exciting diversity of people, which lead to new ideas and new ways of thinking. Both TED speakers show that extraordinary individuals are extraordinary individuals, no matter the circumstances.

EXPLORE MORE

Look at more pictures of Sue Austin's work on the TED Blog at TED.com. Watch other videos of her wheelchair dives. Share what you liked with your class.



Project



Huang Guofu paints at an art studio in Chongqing City, China. Huang, a physically disabled artist, uses paintbrushes held in his mouth and foot.

Project

A. Work with a partner. You are going to research another person whose accomplishments challenge our preconceptions about disability. Choose one of the people below or choose your own person to research, and answer the questions.

- Sudha Chandran
- Lenin Moreno
- Stephen Hawking
- Huang Guofu

1. What is the person's "disability"?
2. What are this person's accomplishments or talents?
3. How is the person's "disability" a challenge?
4. How is the "disability" a strength?

Project

- B. Use your information to create a two-minute presentation. You can use maps, photos, and video to explain your information.**
- C. Give your presentations. Which person would you most like to talk with and why? What questions would you ask him or her?**

EXPLORE MORE

Watch a TED Talk by Stella Young, Aimee Mullins, Chris Downey, or Maysoon Zayid. How is their perspective on disability similar to or different from the views shared by Grandin and Austin?

Part 1

. . . I started using a wheelchair 16 years ago when an extended illness changed the way I could access the world. When I started using the wheelchair, it was a tremendous new freedom. I'd seen my life slip away and become restricted. It was like having an enormous new toy. I could whiz around and feel the wind in my face again. Just being out on the street was exhilarating.

But even though I had this newfound joy and freedom, people's reaction completely changed towards me. It was as if they couldn't see me anymore, as if an invisibility cloak had descended. They seemed to see me in terms of their assumptions of what it must be like to be in a wheelchair. When I asked people their associations with the wheelchair, they used words like *limitation*, *fear*, *pity*, and *restriction*. I realized I'd internalized these responses and it had changed who I was on a core level. A part of me had become alienated from myself. I was seeing myself not from my perspective, but vividly and continuously from the perspective of other people's responses to me.

As a result, I knew I needed to make my own stories about this experience, new narratives to reclaim my identity.

Part 2

I started making work that aimed to communicate something of the joy and freedom I felt when using a wheelchair—a power chair—to negotiate the world. I was working to transform these internalized responses, to transform the preconceptions that had so shaped my identity when I started using a wheelchair, by creating unexpected images. The wheelchair became an object to paint and play with. When I literally started leaving traces of my joy and freedom, it was exciting to see the interested and surprised responses from people. It seemed to open up new perspectives, and therein lay the paradigm shift. It showed that an arts practice can remake one's identity and transform preconceptions by revisioning the familiar.

So when I began to dive, in 2005, I realized scuba gear extends your range of activity in just the same way as a wheelchair does, but the associations attached to scuba gear are ones of excitement and adventure, completely different to people's responses to the wheelchair.

So I thought, “I wonder what'll happen if I put the two together?” [Laughter] And the underwater wheelchair that has resulted has taken me on the most amazing journey over the last seven years.

So to give you an idea of what that's like, I'd like to share with you one of the outcomes from creating this spectacle, and show you what an amazing journey it's taken me on. . . .

It is the most amazing experience, beyond most other things I've experienced in life. I literally have the freedom to move in 360 degrees of space and an ecstatic experience of joy and freedom.

And the incredibly unexpected thing is that other people seem to see and feel that too. Their eyes literally light up, and they say things like, "I want one of those," or, "If you can do that, I can do anything."

And I'm thinking, it's because in that moment of them seeing an object they have no frame of reference for, or so transcends the frames of reference they have with the wheelchair, they have to think in a completely new way. And I think that moment of completely new thought perhaps creates a freedom that spreads to the rest of other people's lives. For me, this means that they're seeing the value of difference, the joy it brings when instead of focusing on loss or limitation, we see and discover the power and joy of seeing the world from exciting new perspectives. For me, the wheelchair becomes a vehicle for transformation. In fact, I now call the underwater wheelchair "Portal," because it's literally pushed me through into a new way of being, into new dimensions and into a new level of consciousness.

And the other thing is, that because nobody's seen or heard of an underwater wheelchair before, and creating this spectacle is about creating new ways of seeing, being and knowing, now you have this concept in your mind. You're all part of the artwork too.